MIL -F-12298 A (MU)
26 October 1965
Superseding
MIL-F-12298 (ORD)
24 October 1952

MILITARY SPECIFICATION

FABRIC, GLASS, WOVEN

1. SCOPE

1.1 Scope - This specification covers the requirements of three weaves of fibrous "E" glass fabric that have been heat cleaned and finished with a methacrylato chromic chloride finish.

1.2 Classification - The fabric shall be of the following types, as specified (see 6.2):

Type I - light-weight

Type II - medium-weight

Type III - heavy-weight

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

CCC-T-191 - Textile, Test Methods
PPP-P-51 - Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cotton, Silks, and Synthetics)

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC MISC

Source: https://assist.dla.mil -- Downloaded: 202640-17T22:40Z
Check the source to verify that this is the current version before use.

2.2 Other publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM D579-64 - Methods of Testing and Tolerances for Woven Glass Fabrics

ASTM D2410-65 - Woven Glass Fabric, Cleaned and After-Finished With Methacrylato Chromic Chloride ("A"-Finish With Volan) For Plastic Laminates

(Copies may be purchased directly from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material - The material shall consist of continuous fibrous "E" glass yarns which are woven to conform to the weave requirement, heat-cleaned, and then finished with a methacrylato chromic chloride finish.

3.2 Width - Glass fabrics shall be furnished in the widths specified by the procuring activity (see 6.2). The acceptable tolerance for fabric width shall be ± 1/4 inch.

3.3 Finish - The amount of finish on the fabric, as measured by the chrome level, shall be not less than 0.03 per cent nor more than 0.06 per cent. Where verification of finish is deemed necessary by the procuring activity, finish shall be determined as specified in 4.6.5.

3.4 Physical properties and construction - The physical properties and construction of the fabric shall conform to the requirements of table I.

Table I - Physical properties and requirements

Property	Type I	Type II	Type III
Yarn (warp and filling)[1/]	450 - 3/2	225 - 1/3	225 - 3/2
Weave	Plain	8-harness satin	8-harness satin
Construction, threads per inch (±2)	34 x 32	57 x 54	54 x 48
Thickness, average, inch	0.008 ± 0.0015	0.0010 ± 0.002	0.020 ± 0.002
Weight, oz./sq. yd. (± 10 per cent)	5.37	8.90	16.75
Breaking strength, minimum			
Warp, pounds	110	145	270
Filling, pounds	90	130	250

[1/] The first number of the yarn designation represents the count number,

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:40Z
Check the source to verify that this is the current version before use.

which is 1/100 of the yardage of raw glass strand per pound. Plies are designated by the two digits following the count number. The first digit shows the number of original strands twisted, and the second, the number of these yarns plied. The total number of strands is the product of both numbers.

3.5 Certification - The manufacturer shall present certification that, (1) the finish on the fabric is a methacrylato chromic chloride, and, (2) the amount of finish on the fabric as measured by the chrome level is not less than 0.03 per cent and no more than 0.06 per cent. The certificate shall be presented prior to, or at the time of delivery of the lot, shall be signed by a responsible agent of the supplier, and shall be accompanied by evidence of the agent's authority to bind his principal.

3.6 Workmanship - The fabric shall be uniform in color and appearance, and shall be free of oil, grease, and dirt spots greater than 2 inches in any direction, holes, cuts, tears, and brittle or fused areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot-A lot shall consist of a maximum of 5000 yards of fibrous glass fabric of the same composition, weave, and width and which is produces by one manufacturer.

4.3 Sampling

4.3.1 Sampling for examination - A sample of closed containers shall be selected from each lot in accordance with Military Standard MIL-STD-105 at Inspection Level I to verify compliance with all stipulations of this specification regarding inspection requirements not involving tests (see 4.5). The sample unit shall be one roll.

4.3.2 Sampling for tests - A sample of sufficient size to prepare the necessary specimens for testing shall be selected at random from each lot. The first convolution of each roll to be sampled shall not be used for test specimens.

4.4 Inspection - Inspection for acceptance of each lot shall consist of determining conformance to the requirements of Section 3, as applicable, and the examination specified in 4.5.

Source: https://assist.dla.mil -- Downloaded: 202640-17T22:40Z
Check the source to verify that this is the current version before use.

4.5 Examination

4.5.1 Visual - Sample units selected in accordance with 4.3.1 shall be examined for the defects listed in table II. The sample shall be placed over a fluorescent panel and visually scanned from a distance of approximately 3 feet. The fluorescent panel shall be a minimum of 3 feet above the floor to insure that no part of the sample will touch the floor and become contaminated.

4.5.2 Width - Sample units shall be examined for the defect and at the acceptable quality level listed in table II.

4.5.3 Packaging, packing and marking - Sample units shall be examined for the defects and at the acceptable quality levels shown in table II.

Table II - Examination

AQL per cent	Classification of defect	Defect	Method of inspection
2.5	Minor 201	Non-uniform in color and appearance	Visual
	Minor 202	Oil or grease spots	Visual
1.5	Major 101	Dirt spots 2 inches or more in any direction	Visual
	Major 102	Hole, cut, or tear where 3 or more yarns are broken at adjoining points	Visual
	Major 103	Brittle or fused area	Visual
2.5	Minor 203	Width not as specified	Approved gage
1.5	Major 104	Improper packaging	Visual
	Major 105	Improper packing	Visual
	Major 106	Lack of, or improper marking	Visual

4.6 Test procedures

4.6.1 Construction - Construction of the fabric shall be determined in accordance with Method 5050.1 of Federal Specification CCC-T-191. The reported value shall be the average of five determinations.

4.6.2 Thickness - Thickness of the fabric shall be determined in accordance with Method 5030 of Federal Specification CCC-T-191 except that the apparatus shall be a dead weight dial micrometer with a 1/4 inch foot weighted so as to give a pressure of 23 to 27 psi. The reported value shall be the average of five determinations.

4.6.3 Weight - Weight of the fabric shall be determined in accordance with Method 5041 of Federal Specification CCC-T-191. The reported value shall be the average of five determinations.

Source: https://assist.dla.mil -- Downloaded: 202640-17T22:40Z
Check the source to verify that this is the current version before use.

4.6.4 Breaking strength - Breaking strength of the fabric shall be determined in accordance with ASTM Specification D579-64, Section 11. Five determinations shall be made in the warp direction and five determinations in the fill direction. The average of the five determinations for each direction shall be reported as the breaking strength in that direction.

4.6.5 Determination of fabric finish - Fabric finish shall be determined by using either of the methods described in 4.6.5.1 and 4.6.5.2.

4.6.5.1 Finish determination using Nesslerimeter - A 10 per cent sodium hydroxide solution shall be prepared by dissolving 100 grams of sodium hydroxide pellets in a beaker containing approximately 500 ml. of distilled water. Pour into a 1 liter volumetric flask and add sufficient distilled water to make a total of 1 liter of solution.

Place several grams of potassium dichromate in an uncovered weighing bottle and heat in an oven for 1 hour at 212°-220°F. Cool in a desiccator for 30 minutes. Weigh 2 samples of the dried dichromate, 0.8484 gram each, in separate 1000 ml. beakers. Dissolve each sample in 200 to 300 ml. of distilled water and transfer the solutions to separate 1 liter volumetric flasks. Add distilled water to bring the volume to the mark. (The solutions will contain 0.0003 grams of chromium per ml. of solution.) Transfer 100 ml. of the standard potassium dichromate stock solution to a 1 liter volumetric flask, and dilute to the mark with distilled water. Use this standard diluted dichromate solution for preparing the standard dichromate working solutions.

Prepare a series of 8 working standards, containing potassium dichromate equivalent to 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, and 0.08 per cent chromium respectively, as follows: Pipette, into a 50 ml. Nessler tube, 2.5 ml. of dichromate diluted stock solution for each 0.01 per cent chromium desired in the working standard. Add 1 ml. of the 10 per cent sodium hydroxide solution and dilute to the mark with distilled water.

Weigh three 1.50 ± 0.01 gram samples of fabric. If necessary cut each sample into 2 or more pieces, so that the fabric will have less bulk and can be covered completely by the liquid. Place each fabric sample in a separate 400 ml. beaker. Add 25 ml. of 10 per cent sodium hydroxide solution, 35 ml. of distilled water, and 1.0 gram of sodium peroxide. Cover each beaker with a watch glass and boil the contents vigorously for 8 minutes, or until frothing ceases. If necessary, add distilled water from a wash bottle to keep the solution from evaporating completely. Cool the beaker's content to 70° ± 2°F. Transfer the contents of each beaker to separate 100 ml. volumetric flask. Carefully rinse all the solution from each beaker into its respective flask with distilled water delivered from a wash bottle. Dilute to the mark and filter using Whatman No. 40 filter paper, or equivalent. Discard the first 15 to 20 ml. of each filtrate.

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:40Z
Check the source to verify that this is the current version before use.

Pour 50 ml. of each filtered solution into a separate Nessler tube, and compare its color with that of the working standard solutions, in a Nesslerimeter. Determine the nearest corresponding chromium percentage.

4.6.5.2 Finish determination using photometer - Photometer measurement of chrome complex content shall be conducted in accordance with paragraph 5c of ASTM Specification D2410.

4.7 Rejection

4.7.1 Examination - A lot shall be rejected when the number of non-conforming units of a sample exceeds the acceptance number for that sample when examined in accordance with 4.5.

4.7.2 Tests - A lot shall be rejected for failure to comply with any of the test requirements of this specification, when tested in accordance with 4.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A, B, and C - Cloth of one type only shall be packaged in accordance with the applicable requirements for Class 2 fabrics of Specification PPP-P-51.

5.2 Packing - Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C - Cloth of one type only shall be packed in accordance with the applicable requirements of Specification PPP-P-51.

5.3 Marking - In addition to any special marking required by the contract or order (see 6.2), shipments shall be marked in accordance with the applicable requirements of Specification PPP-P-51.

6. NOTES

6.1 Intended use - Fabrics covered by this specification are intended for use with phenolic resins in preparing Belleville springs for non-metallic mines.

6.2 Ordering data - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of fabric required (see 1.2).
c. Width of fabric required (see 3.2).
d. Level of packaging required (see 5.1).
e. Level of packing required (see 5.2).
f. Special marking, if required (see 5.3).

Source: https://assist.dla.mil -- Downloaded: 202610-17T22:40Z
Check the source to verify that this is the current version before use.

Custodian:

Army-MU

Preparing activity:

Army-MU

Project 1395-A022

Review activity:

Army-MU

Source: https://assist.dla.mil -- Downloaded: 202640-17T22:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *IF "YES", IN WHAT WAY?*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:40Z
Check the source to verify that this is the current version before use.

Fold

DEPARTMENT OF THE ARMY
Picatinny Arsenal
Dover, New Jersey 07801

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-VP
Dover, New Jersey 07801

Fold

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:40Z
Check the source to verify that this is the current version before use.